SYSTEM REQUIREMENTS SPECIFICATION

Industrial Dual-Motor Controller & Communication Gateway

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Project	Industrial Dual-Motor Controller
Version	V0.1
MCU family	STM32
Authority	System-level product requirements

Important design rule: This document defines what the product shall do. It intentionally does not prescribe most implementation choices. The hardware/firmware designer is responsible for selecting regulators, motor-driver topology, MOSFETs, PHY/transceivers, connector series, PCB stackup, PDN strategy, filtering, thermal solution and component values, provided all requirements and verification criteria are met.

System Requirements — v0.1

1. Product Definition

The product shall be a 24 VDC industrial controller designed to independently control two permanent-magnet DC motors and provide communication with external industrial control systems.

2. Motor Requirements

- Motor quantity: **2**
- Motor type: **Permanent Magnet Brushed DC Motor**
- Motor model: **90ZYTA144-1253**
- Motor nominal voltage: **12 VDC**
- Motor nominal current: **3 A**
- Each motor shall be independently controlled.
- The controller shall provide:
  - Enable/Disable
  - Direction control
  - PWM speed control
  - Current monitoring
  - Fault detection
- Peak/stall current shall be determined during detailed design and verified before final hardware release.

3. Main Power Requirements

- Nominal input voltage: **24 VDC**
- Input voltage range: **20–30 VDC**
- The controller shall generate all required internal power rails from the 24 VDC input.
- Input power protection shall be provided.

- The power architecture shall meet required voltage, current, efficiency, thermal and EMC requirements.

Implementation of power conversion is not fixed.
The designer may select LDO, switching regulator, isolated/non-isolated converter, filtering and protection components based on technical, thermal, EMC, availability and cost requirements.

4. Communication Interfaces

The controller shall provide:

CAN

- CAN interface according to applicable ISO 11898 requirements.
- Configurable communication bitrate.

RS-485

- RS-485 physical interface.
- Differential communication.
- Application protocol shall be defined separately; Modbus RTU may be supported.

Ethernet

- 10/100 Mbps Ethernet.
- Full-duplex operation.
- Ethernet PHY/interface shall be selected by the designer.

USB

- **USB 2.0 Full-Speed**
- Up to 12 Mbps.
- Intended for configuration, diagnostics, firmware update and service.

5. Internal Interfaces

The MCU shall provide interfaces required for internal peripherals, including:

- I^2C
- SPI
- UART
- GPIO
- ADC
- PWM
- Timers
- DMA
- SWD/JTAG or equivalent debug/programming interface

The exact STM32 device shall be selected by the designer based on system requirements.

6. MCU Requirements

- MCU family: **STM32**
- MCU shall provide sufficient:
    - GPIO
    - ADC channels
    - PWM/timers
    - CAN capability/interface
    - UART
    - SPI
    - I²C
    - USB capability
    - Memory
    - Processing performance
- Exact STM32 part number is a **design decision**.

7. Monitoring

The controller shall monitor, where applicable:

- Input voltage
- Logic supply voltage
- Motor current
- Motor-driver fault status
- Board temperature
- Motor-driver temperature
- Communication status

8. Protection

The system shall provide appropriate protection against:

- Reverse polarity
- Over-voltage
- Under-voltage
- Over-current
- Short circuit
- Thermal overload
- Input transients
- ESD
- Electrical disturbances relevant to the intended industrial environment

The exact protection circuit and component selection are the responsibility of the designer.

9. Fault Handling

The controller shall:

- Detect critical faults.
- Disable affected motor output when required.
- Report fault status through the communication interfaces.
- Provide local fault indication.

- Prevent unsafe motor operation following defined fault conditions.
- Support fault recovery/reset according to firmware requirements.

10. Firmware

Firmware shall provide:

- Hardware initialization
- Motor control
- PWM generation
- Current/voltage/temperature monitoring
- Communication handling
- Fault management
- Diagnostics
- Configuration management
- Watchdog supervision
- Firmware update capability

Firmware shall be modular and maintainable.

11. PCB / Hardware Design

The PCB shall be designed for industrial operation and shall consider:

- Power integrity
- Signal integrity
- PDN
- Thermal management
- Current density
- Creepage/clearance where applicable
- EMC/EMI
- Grounding
- Decoupling
- Return-current paths
- Connector robustness
- Manufacturing constraints

PCB stack up, layer count, PDN architecture, decoupling strategy, regulator topology, connector type and individual component selection are design decisions.

12. Thermal

The designer shall evaluate:

- Power dissipation
- Motor-driver losses
- Regulator losses
- MCU and PHY losses
- PCB thermal performance
- Component temperature
- Derating

Thermal simulation and/or calculation shall be performed for critical components.

13. Environmental

Initial design target:

- Industrial indoor environment
- Operating temperature: **TBD** (To Be Determined)
- Storage temperature: **TBD** (To Be Determined)
- Relative humidity: **TBD** (To Be Determined)
- Vibration/shock requirements: **TBD** (To Be Determined)

These values shall be finalized before design freeze.

14. Verification

The final prototype shall be verified for:

- Power-up behavior
- Motor operation
- Motor current
- Communication
- USB
- CAN
- RS-485
- Ethernet
- Protection functions
- Fault handling
- Thermal performance
- Long-duration operation
- EMC pre-compliance

15. Standards / Design References

The design shall consider applicable requirements from:

- ISO 11898 — CAN
- TIA/EIA-485 — RS-485
- IEEE 802.3 — Ethernet
- USB 2.0 specification
- IEC 61000 series — EMC/environmental immunity and emissions as applicable
- Relevant industrial motor-drive/electrical safety standards where applicable
- IPC standards like IPC-610